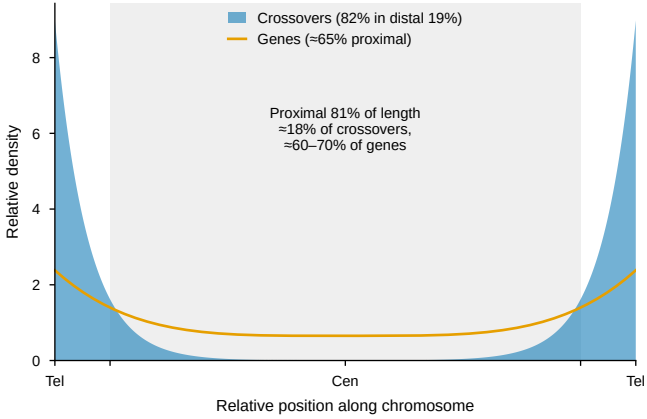
a



b

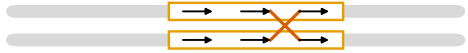
Structural heterozygote



Zhongmai 895 × Yangmai 16, Emai 580 or Avocet S

Crossovers in the 4.1-Mb inverted segment are suppressed or yield unbalanced gametes → no usable recombinants

Structurally matched cross

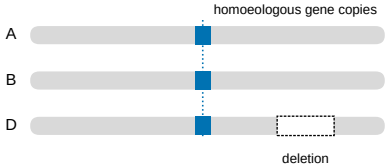


Zhongmai 895 × Jimai 22

Normal recombination; *YR86* mapped to 0.15 cM (710.27–712.56 Mb, Chinese Spring RefSeq)

c

Homoeologous buffering



Dosage from A and B tolerates structural change (wide design space for editing and introgression)

A recessive allele is masked unless all homoeologs are stacked: $P(\text{access}) = P_A \times P_B \times P_D$

Gene-pool accessibility for the D and AB subgenomes

Ae. tauschii	**Tetraploid AB relatives**	**Wild relatives (e.g. *Ae. mutica*)**	**Engineered structure**
D-genome donor; homologous with D	Homologous with A and B only	Homoeologous only; suppressed by *Ph1*/*Ph2*	Inversion reversal, centromere editing, large-fragment edits
Synthetic hexaploids, direct crosses	Bridge crosses	*zip4-5B*, *msh7-3D* or *Ph1*-suppressor donors	Verify fertility and transmission

Native recombination with wheat chromosomes decreases; dependence on meiotic or structural intervention increases